

Homework3

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```
library(tidyverse)
library(here)
library(janitor)
library(readxl)

kelp <- read_csv(here("data", "temp-kelp.csv"))

my_data <- read_excel(here("data", "my_data.xlsx")) |> clean_names()
```

Problem 1

a. The two appropriate tests would be a correlation test and a linear regression. A correlation test measures the strength and direction of the relationship between ocean temperature ($^{\circ}\text{C}$) and giant kelp frond elongation rate (cm day^{-1}), while a linear regression examines how changes in ocean temperature predict changes in kelp growth rate. Correlation only measures association between the two variables, but regression models the relationship and identifies temperature as the explanatory variable and elongation rate as the response variable.

b.

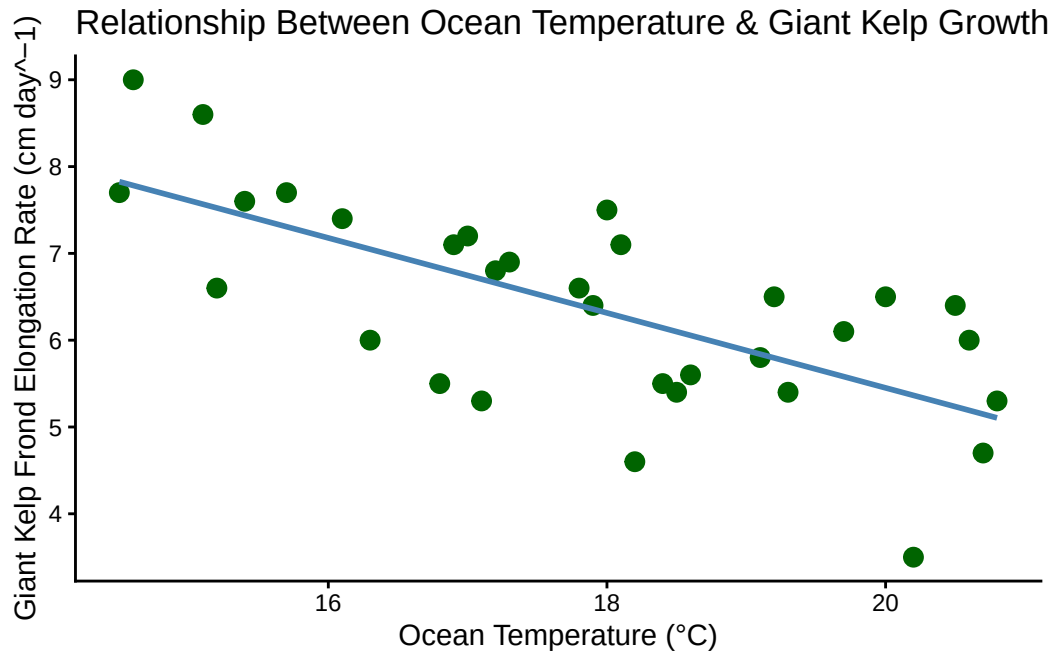
```
ggplot(kelp, aes(
  x = temp_c,
  y = kelp_elong)) +

  geom_point(color = "darkgreen", size = 3) +

  geom_smooth(method = "lm", color = "steelblue", se = FALSE) +

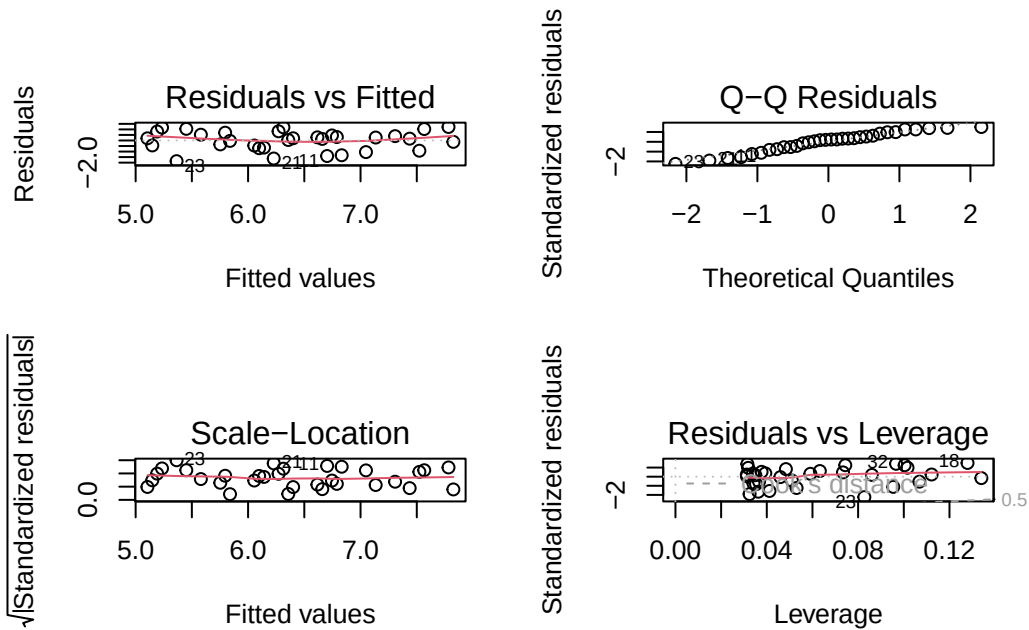
  labs(x = "Ocean Temperature ( $^{\circ}\text{C}$ )",
       y = "Giant Kelp Frond Elongation Rate ( $\text{cm day}^{-1}$ )",
```

```
title = "Relationship Between Ocean Temperature & Giant Kelp Growth") +  
theme_classic()
```



c. To evaluate the relationship between ocean temperature and giant kelp frond elongation rate, I checked the assumptions for a linear regression. Specifically, I checked for linearity, normality of residuals, and equal variance of residuals. I assessed these assumptions using a scatterplot with a regression line and diagnostic plots from the regression model. Based on the plots, the assumptions appeared reasonably met because the relationship looked linear and the residuals did not show major deviations from normality or equal variance.

```
# Create diagnostic plots  
  
kelp_lm <- lm(kelp_elong ~ temp_c,  
             data = kelp)  
  
par(mfrow = c(2, 2))  
plot(kelp_lm)
```



```
# Display the regression results
summary(kelp_lm)
```

Call:

```
lm(formula = kelp_elong ~ temp_c, data = kelp)
```

Residuals:

Min	1Q	Median	3Q	Max
-1.8643	-0.5017	0.1790	0.5659	1.2177

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.08645	1.49219	9.440	1.72e-10 ***
temp_c	-0.43179	0.08321	-5.189	1.37e-05 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.8665 on 30 degrees of freedom

Multiple R-squared: 0.473, Adjusted R-squared: 0.4554

F-statistic: 26.93 on 1 and 30 DF, p-value: 1.366e-05

d. To evaluate the strength of the relationship between ocean temperature and giant kelp

frond elongation rate, I used a linear regression because both variables are continuous and I was interested in determining whether temperature could predict kelp growth rate.

The linear regression showed a relationship between ocean temperature and giant kelp frond elongation rate (report the slope coefficient, R^2 , and p-value from your output here). This suggests that as ocean temperature changes, kelp elongation rate also changes in a predictable way.

e. The results of this test suggest that ocean temperature is related to giant kelp frond elongation rate, meaning that changes in temperature may influence how quickly giant kelp grows. If warmer temperatures are associated with reduced growth rates, this could indicate that continued ocean warming from climate change may negatively impact giant kelp forest health and productivity. These findings highlight the importance of monitoring ocean temperature trends and protecting kelp ecosystems that may be vulnerable to warming conditions.

f.

```
# Run a correlation test  
  
cor.test(kelp$temp_c, kelp$kelp_elong)
```

Pearson's product-moment correlation

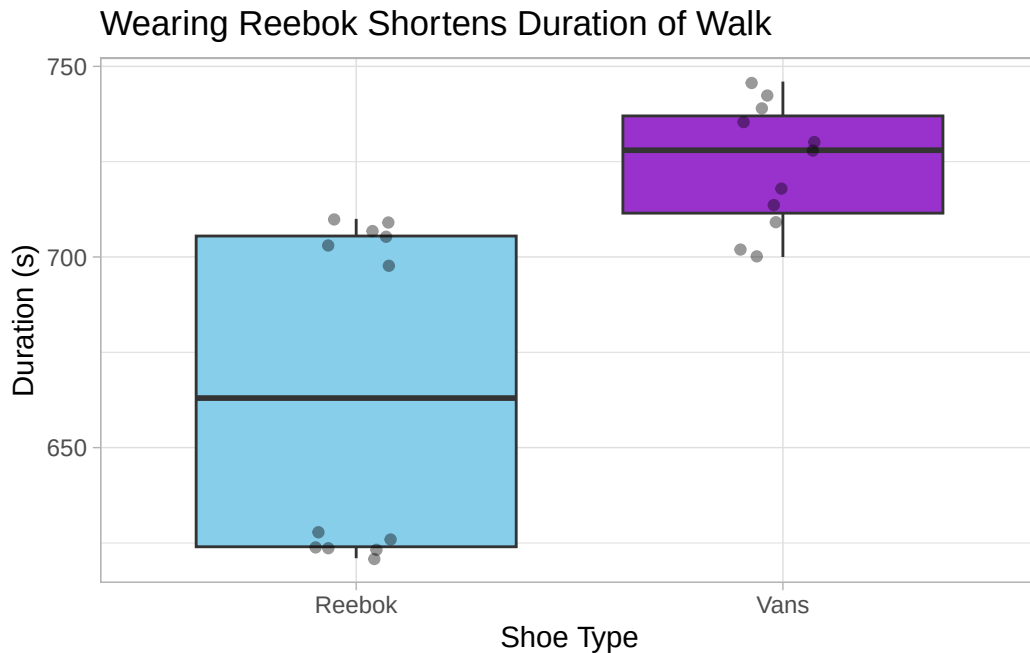
```
data: kelp$temp_c and kelp$kelp_elong  
t = -5.189, df = 30, p-value = 1.366e-05  
alternative hypothesis: true correlation is not equal to 0  
95 percent confidence interval:  
-0.8359665 -0.4460157  
sample estimates:  
cor  
-0.6877489
```

The correlation test and the linear regression would most likely lead to the same decision about the null hypothesis because both tests evaluate the relationship between ocean temperature and giant kelp frond elongation rate. However, the correlation test measures the strength and direction of the association between the two continuous variables using the correlation coefficient (r), while the linear regression additionally models how temperature predicts changes in elongation rate through the regression slope. As a result, both tests should provide similar explanations about whether the relationship is positive or negative, and statistically significant.

Problem 2. Personal data

a.

```
# base layer: ggplot using my_data
ggplot(data = my_data,
aes(x = shoe_type, # maps shoe type to x-axis
y = duration_s, # maps duration to y-axis
fill = shoe_type # fills boxplots with color based on shoe type
)) +
geom_boxplot() + #second layer: boxplot
# third layer: jitterplot
geom_jitter(
width = 0.1, # spreads points horizontal
alpha = 0.4 # makes points semi-transparent
) +
scale_fill_manual(values = c( #customizes colors
"Reebok" = "skyblue",
"Vans" = "darkorchid"
)) +
labs( # label x- and y- axes along with the title
x = "Shoe Type",
y = "Duration (s)",
title = "Wearing Reebok Shortens Duration of Walk"
) +
theme_light() + # apply different theme
theme(legend.position = "none") # remove legend
```



b. Figure 1. Boxplots and jittered data points comparing walking duration, in seconds, between wearing Reebok and Vans shoes. The boxplots display the median, interquartile range, and overall spread of walking durations for each shoe type, while the jittered points show individual observations. Reebok shoes generally showed shorter walking durations and less variation compared to Vans shoes. Colors were used to distinguish shoe types, with Reebok represented in blue and Vans represented in purple.

Problem 3. Affective visualization

a. An affective visualization of my data about walking to work in different shoes could represent both the duration of each walk and how each pair of shoes felt to wear. Instead of only showing numerical differences, the visualization could use simple visual elements like color or shape to represent comfort level or effort. More comfortable shoes could be shown with lighter or more consistent visuals, while less comfortable shoes could be represented with darker or more irregular patterns. This would help show not just the measured walking times, but also the overall experience associated with each shoe type.

b.

```
knitr::include_graphics(here::here("data", "visualization.png"))
```

Walking Data
Vans vs Reebok

Shoe

[illegible]

Traffic

● = low
 😊 = med
 ●●● = high

